S EC2 Windows Instance Setup

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Repository: Ayusha-hub/CSA--Unit4_assignment

1. Objective

Launch a Windows Server instance on Amazon EC2 and connect to it with Remote Desktop. The instance sits in a public subnet, keeps a stable address with an Elastic IP, and accepts RDP only through a security group rule on port 3389. The Administrator password is decrypted at connection time and is not written in this report.

2. Setup steps

1. **Create the VPC and public subnet.** Create a VPC, attach an internet gateway, and add a public subnet whose route table sends internet traffic to that gateway.
2. **Configure the security group.** Allow inbound RDP, which is TCP port 3389, from the administrator's own IP address. Leave the private key and the Windows password out of the security group.
3. **Launch the EC2 Windows instance.** Choose a Windows Server image, place the instance in the public subnet, attach the security group, and select a key pair. The private key stays on the administrator's computer.
4. **Assign an Elastic IP.** Allocate an Elastic IP and associate it with the instance. Remote Desktop uses this address, so the target does not change when the instance is stopped and started.
5. **Connect with RDP.** In the EC2 console, decrypt the Windows password with the key pair. Open Remote Desktop to the Elastic IP as the user Administrator. Use the decrypted password only in the Remote Desktop client. Do not save it in the repository or in this report.

3. Architecture

The path is the one recorded in the project guide: the user laptop opens RDP to the Elastic IP, and that address forwards to the Windows instance inside the public subnet.

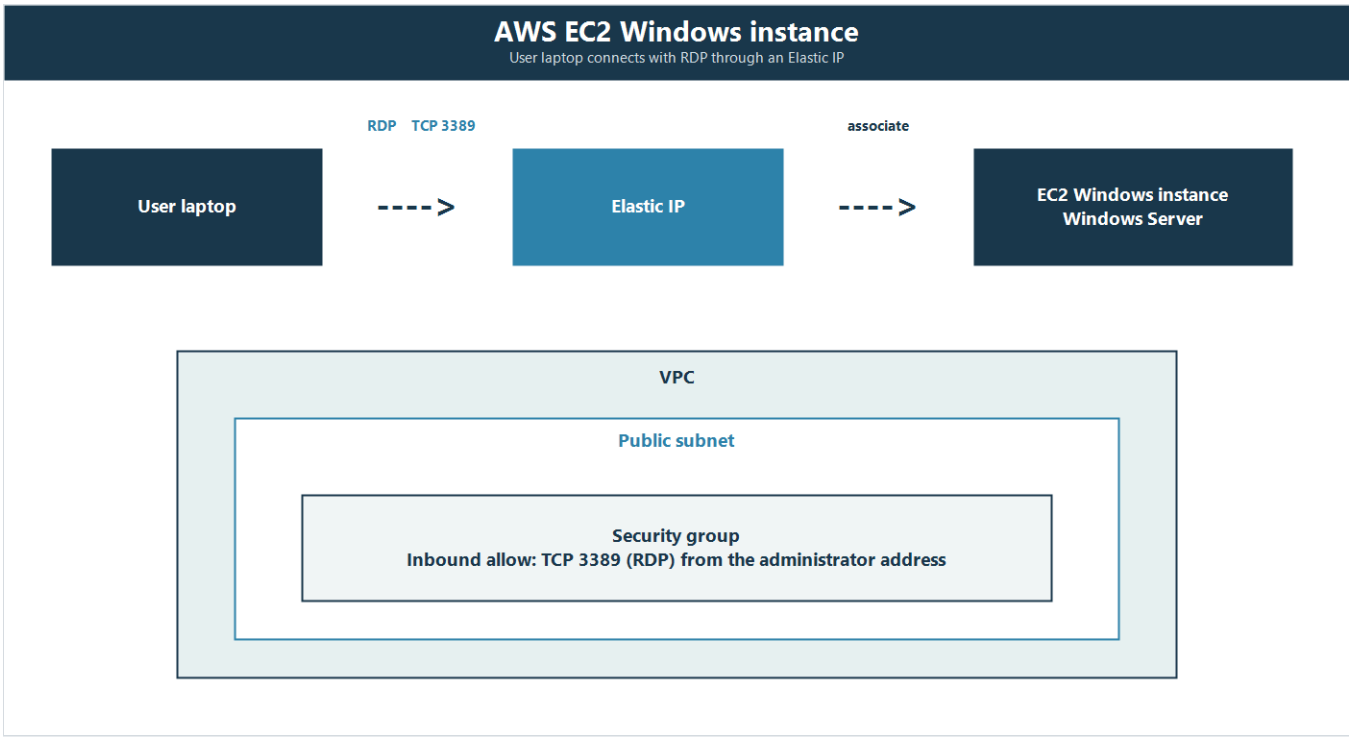


Figure 1. User laptop, Elastic IP, Windows instance, and the VPC path.

4. Connection details

These are the connection settings. The password is excluded.

Item	Value
Protocol	Remote Desktop (RDP)
Port	3389
Target	Elastic IP associated with the Windows instance
Username	Administrator
Password	Excluded. Decrypt it in the EC2 console with the key pair, then type it only into the Remote Desktop client.
Key pair	Used to decrypt the password. The private key is not stored in this repository.

5. Security configuration

RDP is the only administrative path this setup opens. The security group is the control that allows or blocks that path.

Security group for RDP

Direction	Inbound
Type	RDP
Protocol	TCP
Port	3389
Source	Administrator IP address
Password	Not stored in this report

Figure 2. Inbound security group rule. The password is not part of the configuration.

- Inbound: TCP 3389 from the administrator’s IP address, so Remote Desktop is not left open to the whole internet.
- The instance is launched in a public subnet so the Elastic IP can be reached.
- The private key and the decrypted Administrator password stay off the instance record in this report.

6. Conclusion

The Windows instance is reached with RDP on port 3389 through an Elastic IP. A security group allows that port from the administrator’s address, and the instance remains in a public subnet inside the VPC. The Administrator password is decrypted with the key pair when the connection is made, and it is intentionally left out of the repository and this report.